

Linear Programming: Midterm Review

Lecture skeleton, following the [unit lecture notes](#). Use the watershed restoration example below throughout; add solution walkthroughs after revising the outline.

1 LP Overview

- What makes a problem a linear program?

1.1 LP assumptions

- **Objective-function appropriateness:** one stated objective is sufficient to rank feasible choices.
- **Decision-variable appropriateness:** all relevant choices are included and controlled by the decision maker.
- **Constraint appropriateness:** the constraints accurately represent the relevant limits, requirements, and units.
- **Proportionality:** each activity's per-unit contribution and resource use do not change with its level.
- **Additivity:** the contribution of one activity does not depend on the level of another activity.
- **Divisibility:** decision variables may take fractional values.
- **Certainty:** objective, technical, and right-hand-side coefficients are known fixed values.

2 Formulating an LP

- Define decision variables and units; identify the objective, constraints, and sign restrictions.
- Translate between scalar and matrix forms: identify x , c , A , and b .

2.1 Example: Watershed Restoration

A watershed district wants to meet annual sediment and phosphorus reduction targets at the lowest annual cost. It can establish vegetated buffer strips or restore wetlands. For this hypothetical example, assume constant reductions and costs per acre:

Practice	Annual cost (dollars/acre)	Sediment reduction (tons/acre/year)	Phosphorus reduction (kg/acre/year)
Buffer strips	400	6	2
Restored wetlands	600	3	6

The district must reduce sediment by at least 120 tons per year and phosphorus by at least 90 kg per year. Only 16 acres are suitable for wetland restoration. Assume that buffer-strip acreage has no additional upper limit, that the two practices occupy separate sites, and that fractional acres are allowed.

Formulation prompt. Define the choices and their units, write the objective and all constraints, and identify which constraints represent minimum requirements and which represent resource limits.

2.1.1 Scalar form

Let B be acres of buffer strips and W be acres of restored wetlands. Annual cost is

$$\begin{aligned} \min_{B,W} \quad & 400B + 600W \\ \text{s.t.} \quad & 6B + 3W \geq 120 \quad (\text{sediment reduction}), \\ & 2B + 6W \geq 90 \quad (\text{phosphorus reduction}), \\ & W \leq 16 \quad (\text{suitable wetland acreage}), \\ & B, W \geq 0. \end{aligned}$$

The first two constraints require **at least** the target reductions; the third permits **at most** the available wetland acreage. Each constraint compares quantities with matching units, and the objective is measured in dollars per year.

2.1.2 Matrix form

To write all structural constraints with the same inequality direction, multiply $W \leq 16$ by -1 , giving $-W \geq -16$. Then

$$\begin{aligned} \min_{x \geq 0} \quad & c^\top x \quad \text{s.t.} \quad Ax \geq b, \\ x = \begin{bmatrix} B \\ W \end{bmatrix}, \quad c = \begin{bmatrix} 400 \\ 600 \end{bmatrix}, \quad A = \begin{bmatrix} 6 & 3 \\ 2 & 6 \\ 0 & -1 \end{bmatrix}, \quad b = \begin{bmatrix} 120 \\ 90 \\ -16 \end{bmatrix}. \end{aligned}$$

Why would replacing the first two inequalities with equalities change the model? Which LP assumptions justify adding the reductions across practices and allowing fractional acres?

3 Graphical and Algebraic Solutions

3.1 Plot the constraints and feasible region

Place buffer-strip acreage B on the horizontal axis and wetland acreage W on the vertical axis. Rewrite the reduction requirements as

$$W \geq 40 - 2B \quad (\text{sediment}), \quad W \geq 15 - \frac{1}{3}B \quad (\text{phosphorus}).$$

The feasible region lies **above both reduction lines**, **below** $W = 16$, and in the nonnegative quadrant. The sediment boundary has intercepts $(20, 0)$ and $(0, 40)$; the phosphorus boundary has intercepts $(45, 0)$ and $(0, 15)$.

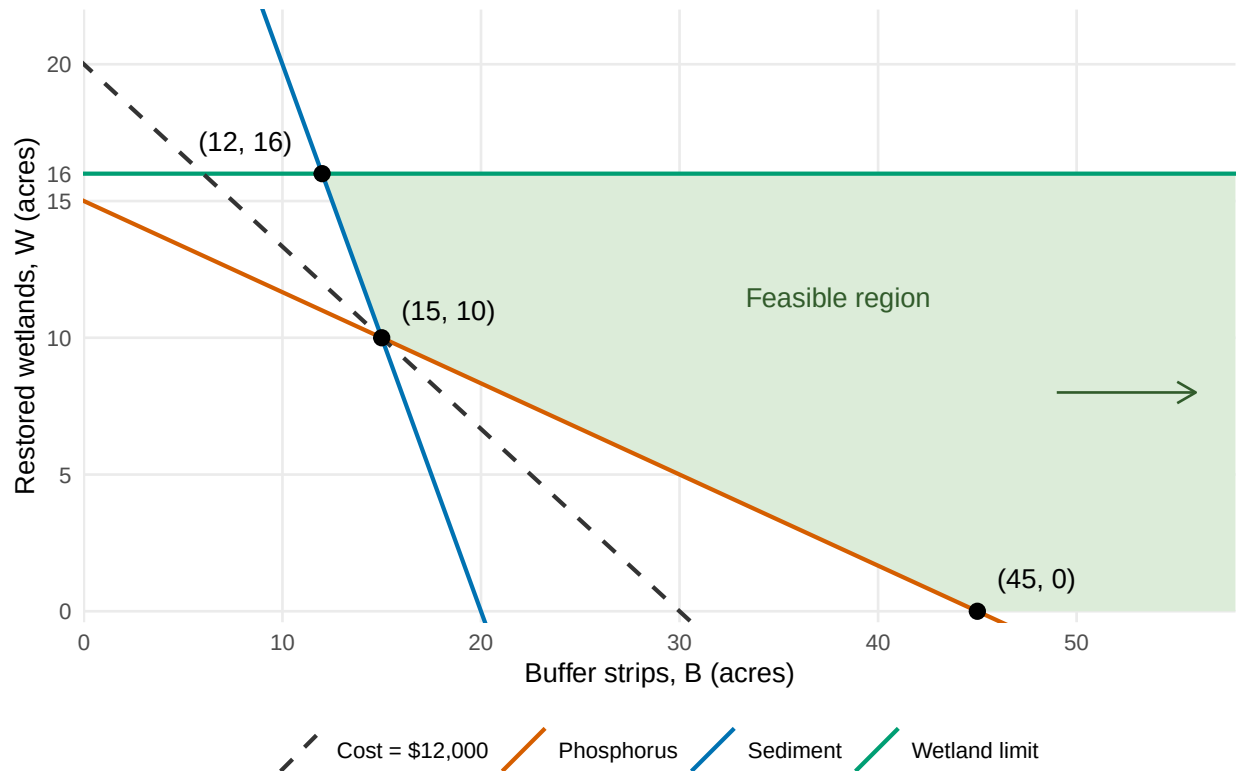


Figure 1: Watershed restoration LP. The shaded feasible region continues to the right without bound. The sediment line's vertical intercept, $(0, 40)$, is above the plotted window. The dashed line represents annual cost of \$12,000.

The feasible region is **convex**: a line segment between any two feasible plans remains feasible. It is also **unbounded** because buffer acreage has no upper limit. An unbounded feasible region does not imply an unbounded objective: adding unnecessary buffer acreage increases cost, so it cannot improve this minimization problem.

3.2 Plot the objective function

Graphically, a line of constant cost Z is

$$W = \frac{Z}{600} - \frac{2}{3}B.$$

Move this line toward the origin to lower cost. The lowest-cost line that still touches the feasible region passes through $(15, 10)$. Its slope, $-2/3$, lies strictly between the slopes of the two reduction boundaries, -2 and $-1/3$, so this optimum is unique. If a supporting cost line instead coincided with a feasible edge, every point on that edge would be optimal.

3.3 Find the vertices algebraically

In two dimensions, find candidate vertices by setting two independent constraint boundaries to equality. Nonnegativity boundaries count too. Then check each candidate against **every** constraint; an intersection alone is not enough.

3.3.1 Sediment requirement and wetland limit

Set $W = 16$ and make the sediment constraint bind:

$$6B + 3(16) = 120 \quad \Rightarrow \quad 6B = 72 \quad \Rightarrow \quad B = 12.$$

At $(B, W) = (12, 16)$, phosphorus reduction is $2(12) + 6(16) = 120 \geq 90$. Both acreages are nonnegative, so $(12, 16)$ is a feasible vertex.

3.3.2 Sediment and phosphorus requirements

Set both reduction constraints to equality and simplify:

$$\begin{aligned} 6B + 3W &= 120 & \Rightarrow & \quad 2B + W = 40, \\ 2B + 6W &= 90. \end{aligned}$$

Subtract the first simplified equation from the second:

$$5W = 50 \quad \Rightarrow \quad W = 10, \quad 2B + 10 = 40 \quad \Rightarrow \quad B = 15.$$

Both acreages are nonnegative and $10 \leq 16$, so $(15, 10)$ is a feasible vertex.

3.3.3 Phosphorus requirement and zero wetlands

Set $W = 0$ and make the phosphorus constraint bind:

$$2B + 6(0) = 90 \quad \Rightarrow \quad B = 45.$$

Sediment reduction is $6(45) + 3(0) = 270 \geq 120$, and $0 \leq 16$. Thus $(45, 0)$ is a feasible vertex. Here the nonnegativity constraint on wetlands supplies one of the two binding boundaries.

4 Simplex Intuition

- **Start at a feasible vertex.** Suppose simplex starts at $(B, W) = (45, 0)$, with annual cost $400(45) = \$18,000$. The origin is infeasible because it meets neither reduction target; a solver first needs a feasible starting basis (often found through a preliminary phase called Phase I).
- **Choose an improving direction.** Let wetlands enter the solution while the phosphorus requirement remains binding. Since $2B + 6W = 90$, each additional acre of wetlands replaces three acres of buffers. With $W = t$,

$$B = 45 - 3t, \quad Z = 400(45 - 3t) + 600t = 18,000 - 600t.$$

Cost falls by \$600 for each acre of wetlands added along this edge.

- **Move until another constraint binds.** Along this edge, sediment reduction is $6(45 - 3t) + 3t = 270 - 15t$. Feasibility requires

$$t \leq 10 \quad (\text{sediment}), \quad t \leq 16 \quad (\text{wetland limit}), \quad t \leq 15 \quad (\text{nonnegative buffers}).$$

The smallest limit is $t = 10$: the sediment requirement binds first. This is the **ratio-test logic**. The pivot reaches $(15, 10)$ at a cost of \$12,000; wetlands enter the basis and the sediment surplus variable leaves it.

- **Check whether another move improves cost.** Returning toward (45, 0) raises cost. Moving toward (12, 16) along the sediment boundary replaces only half an acre of buffers per added wetland acre, raising cost by $600 - 400(1/2) = \$400$ per acre. Neither adjacent edge offers an improvement, so simplex stops at (15, 10).
- **General idea.** Simplex uses this same sequence in larger problems: choose an improving entering variable, use the ratio test to preserve feasibility, pivot to a neighboring basic feasible solution, and check for optimality. This example needs just one improving pivot from the chosen starting vertex.

5 Sensitivity Analysis and Economic Interpretation

5.1 Objective-coefficient sensitivity

Let c_B and c_W denote annual costs per acre. Changing these coefficients rotates the cost line but leaves the feasible region unchanged:

$$Z = c_B B + c_W W, \quad W = \frac{Z}{c_W} - \frac{c_B}{c_W} B.$$

For positive costs, the current vertex stays optimal when the cost-line slope lies between the slopes of the two binding boundaries:

$$-2 \leq -\frac{c_B}{c_W} \leq -\frac{1}{3} \quad \Longleftrightarrow \quad \frac{1}{3} \leq \frac{c_B}{c_W} \leq 2.$$

- **Buffer cost:** hold $c_W = 600$. Then $1/3 \leq c_B/600 \leq 2$, so $200 \leq c_B \leq 1200$.
- **Wetland cost:** hold $c_B = 400$. Then $1/3 \leq 400/c_W \leq 2$, so $200 \leq c_W \leq 1200$.

Annual cost coefficient	Current	Allowable decrease	Allowable increase
Buffers, c_B (dollars/acre)	400	200	800
Wetlands, c_W (dollars/acre)	600	400	600

Inside these ranges, the unique optimal plan remains (15, 10), but its cost changes.

At a range endpoint, the current plan remains optimal but is no longer unique:

- If $c_B/c_W = 1/3$, the cost line coincides with the phosphorus edge between (15, 10) and (45, 0).
- If $c_B/c_W = 2$, it coincides with the sediment edge between (15, 10) and (12, 16).

Every point on the corresponding edge has the same cost. Beyond an endpoint, another vertex becomes better. If both costs change, check their **ratio** rather than applying the two one-at-a-time ranges independently.

5.2 RHS ranging

Let S be the sediment target (tons/year), P the phosphorus target (kg/year), and H the wetland acreage limit. Keep the original costs, $c_B = 400$ and $c_W = 600$. As long as the current basis remains optimal, both reduction requirements bind:

$$\begin{aligned} 6B + 3W &= S, \\ 2B + 6W &= P. \end{aligned} \quad \Rightarrow \quad B = \frac{2S - P}{10}, \quad W = \frac{3P - S}{15}.$$

The basic variables are B , W , and unused wetland capacity $H - W$. RHS changes leave the objective and technical coefficients unchanged, so the current basis remains optimal as long as these three quantities stay nonnegative.

5.2.1 Range for the sediment target

Hold $P = 90$ and $H = 16$. Substituting gives

$$B = \frac{S - 45}{5}, \quad W = 18 - \frac{S}{15}.$$

Check all three feasibility conditions:

$$\begin{aligned} B \geq 0 &\Rightarrow S \geq 45, \\ W \geq 0 &\Rightarrow S \leq 270, \\ W \leq 16 &\Rightarrow S \geq 30. \end{aligned}$$

The tightest bounds give $45 \leq S \leq 270$. From the baseline target of 120, the allowable decrease is 75 tons/year and the allowable increase is 150 tons/year. At the lower endpoint buffers reach zero; at the upper endpoint wetlands reach zero.

5.2.2 Ranges for phosphorus and wetland capacity

Hold $S = 120$ and $H = 16$ while changing P :

$$B = 24 - \frac{P}{10}, \quad W = \frac{P}{5} - 8.$$

Nonnegative buffers require $P \leq 240$; nonnegative wetlands require $P \geq 40$; and $W \leq 16$ requires $P \leq 120$. Together these give $40 \leq P \leq 120$. At the lower endpoint wetlands reach zero; at the upper endpoint the wetland cap binds.

For changes to H alone, the reduction targets still imply $(B, W) = (15, 10)$. Unused capacity is $H - 10$, so the range is $H \geq 10$. The district can lose six acres of suitable wetland capacity before the plan must change.

RHS parameter	Current	Range for current basis	Allowable decrease	Allowable increase
Sediment, S (tons/year)	120	[45, 270]	75	150
Phosphorus, P (kg/year)	90	[40, 120]	50	30
Wetland limit, H (acres)	16	$[10, \infty)$	6	∞

These are one-at-a-time ranges. With simultaneous RHS changes, check $B = (2S - P)/10 \geq 0$, $W = (3P - S)/15 \geq 0$, and $W \leq H$ directly.

5.3 Shadow prices: the marginal cost of a tighter requirement

Substitute the optimal acreages into the objective:

$$\begin{aligned} Z^*(S, P, H) &= 400 \left(\frac{2S - P}{10} \right) + 600 \left(\frac{3P - S}{15} \right) \\ &= 40S + 80P. \end{aligned}$$

This expression holds while the current basis remains optimal. Its slopes are the **shadow prices**, measured as the change in minimum annual cost per unit increase in the original RHS parameter:

RHS change	Change in minimum annual cost	Interpretation
Sediment target rises by 1 ton/year	+\$40	Additional required abatement raises cost.
Phosphorus target rises by 1 kg/year	+\$80	Additional required abatement raises cost.
Wetland cap rises by 1 acre	\$0	Extra capacity is unused at the baseline.

The positive target prices measure the **cost of stricter requirements**, not the environmental benefits of reducing pollution. Increasing a capacity limit relaxes a constraint, so its effect on minimum cost is nonpositive; the value of extra capacity is the resulting **cost saving**.

5.3.1 Use the ranges before using the prices

- **Raise the sediment target from 120 to 150.** This is within its range. The plan becomes $B = (150 - 45)/5 = 21$, $W = 18 - 150/15 = 8$, and annual cost rises by $40(30) = \$1,200$ to \$13,200. The basis stays the same even though acreages change.
- **Raise the phosphorus target from 90 to 100.** The new plan is $(14, 12)$, and annual cost rises by $80(10) = \$800$ to \$12,800.
- **Raise the phosphorus target from 90 to 130.** This exceeds the range. At $P = 120$, wetlands reach their 16-acre cap and annual cost is \$14,400. Beyond that point, each additional kg/year must come from buffers: half an acre costs \$200. At $P = 130$, $W = 16$ and $2B + 6(16) = 130$ imply $B = 17$, with annual cost $400(17) + 600(16) = \$16,400$. Applying \$80 to all 40 additional units would understate cost.
- **Acquire more suitable wetland acreage.** Raising the baseline cap from 16 to 17 saves nothing, so the district should not pay for it solely to reduce this model's restoration costs. If capacity instead falls to 8 acres, re-solving gives $(B, W) = (21, 8)$ and cost \$13,200. Increasing that cap from 8 to 9 gives $(18, 9)$ and cost \$12,600: the extra acre now saves \$600 per year. An annual access fee below \$600 would be worthwhile for that acre, holding restoration costs fixed.

At a range endpoint, the marginal cost can differ on either side. A shadow price is valid for changes within its range, not indefinitely. Objective-coefficient ranges preserve the current optimal acreages; RHS ranges preserve the current basis and shadow prices, while acreages may change.

6 Duality and Complementary Slackness

6.1 Economic motivation: hire a contractor

Suppose the district considers paying someone else to take over its cleanup obligations. The economic question is:

What is the most the district should pay a contractor to meet its pollution-reduction targets, if the district also gives the contractor access to its suitable wetland land?

Imagine the contractor submits an annual bill with:

- A charge y_S for each ton of required sediment reduction per year.
- A charge y_P for each kilogram of required phosphorus reduction per year.
- A credit y_H for each acre of wetland capacity the district supplies.

The total annual bill is

$$120y_S + 90y_P - 16y_H.$$

The district can challenge the contractor's prices using its own restoration options: "An acre of buffers supplies six tons of sediment reduction and two kilograms of phosphorus reduction per year for \$400. You cannot justify charging more than \$400 for that package." Thus,

$$6y_S + 2y_P \leq 400.$$

The wetland option imposes the analogous restriction, including the land-capacity credit: $3y_S + 6y_P - y_H \leq 600$.

The dual finds the largest contractor bill consistent with those alternatives. At the optimal rates (40, 80, 0), that bill is \$12,000, exactly what the district would spend doing the work itself.

6.2 Write the dual

For the minimization form used above, the primal and dual are

$$\underbrace{\min_{x \geq 0} c^\top x \quad \text{s.t.} \quad Ax \geq b}_{\text{primal}} \quad \longleftrightarrow \quad \underbrace{\max_{y \geq 0} b^\top y \quad \text{s.t.} \quad A^\top y \leq c}_{\text{dual}}.$$

Introduce one dual variable per primal constraint: y_S for sediment reduction, y_P for phosphorus reduction, and y_H for the transformed wetland constraint $-W \geq -16$. Each primal activity supplies one dual constraint:

$$\begin{aligned} \max_{y_S, y_P, y_H} \quad & 120y_S + 90y_P - 16y_H \\ \text{s.t.} \quad & 6y_S + 2y_P \leq 400 \quad (\text{buffers}), \\ & 3y_S + 6y_P - y_H \leq 600 \quad (\text{wetlands}), \\ & y_S, y_P, y_H \geq 0. \end{aligned}$$

The negative signs on y_H come from multiplying the wetland cap by -1 . y_S and y_P value additional required pollution reduction. y_H measures the annual cost saving from an extra acre of wetland capacity; the shadow price with respect to the **original cap** H is $-y_H$.

6.3 Find the dual solution using complementary slackness

Start from the primal solution $(B, W) = (15, 10)$. The wetland cap has six acres of slack, so complementary slackness requires

$$y_H(16 - W) = 0 \quad \Rightarrow \quad 6y_H = 0 \quad \Rightarrow \quad y_H = 0.$$

Both activities are positive, so both dual constraints must bind:

$$\begin{aligned} 6y_S + 2y_P &= 400, \\ 3y_S + 6y_P &= 600. \end{aligned}$$

Divide the first equation by two and subtract it from the second:

$$5y_P = 400 \quad \Rightarrow \quad y_P = 80, \quad 3y_S + 80 = 200 \quad \Rightarrow \quad y_S = 40.$$

Thus the candidate dual solution is $(y_S, y_P, y_H) = (40, 80, 0)$, matching the shadow prices obtained from RHS sensitivity. It is dual feasible: all variables are nonnegative, $6(40) + 2(80) = 400$, and $3(40) + 6(80) - 0 = 600$.

The remaining complementary slackness conditions also hold: the sediment and phosphorus requirements have zero surplus, so $y_S(6B + 3W - 120) = 0$ and $y_P(2B + 6W - 90) = 0$.

6.4 Verify strong duality

For any primal-feasible x and dual-feasible y , **weak duality** gives

$$b^\top y \leq y^\top Ax \leq c^\top x.$$

Thus any feasible dual value is a lower bound on the cost of every feasible restoration plan. Evaluate both objectives at our feasible solutions:

$$\begin{aligned} \text{Primal cost: } & 400(15) + 600(10) = \$12,000, \\ \text{Dual value: } & 120(40) + 90(80) - 16(0) = \$12,000. \end{aligned}$$

The feasible plan costs exactly as much as the dual lower bound, proving that **both solutions are optimal and the duality gap is zero**. This verifies strong duality. Economically, the targets valued at their shadow prices exactly account for minimum restoration cost; unused wetland capacity has no scarcity value.